Vijay Roshan S

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Summary

Computer Science graduate (2026) with production internship experience at Ford Motor Company building LLM-powered developer tools deployed across global engineering teams. Skilled in Python, Java, deep learning (TensorFlow, PyTorch), and full-stack web development (React, FastAPI). Seeking Software / ML Engineer roles where system design and measurable product impact matter.

Education

Madras Institute of Technology – Anna University — Chennai, India
B.E. Computer Science and Engineering — *2022 – 2026*
- Relevant coursework: Data Structures & Algorithms, DBMS, OS, Computer Networks, Machine Learning, OOP, Java

Sri Sankara Senior Secondary School (CBSE) — Chennai, India
Class XII (HSC) – 92% — *2022*
Class X (SSLC) – 89% — *2020*

Experience

Ford Motor Company — Chennai, India
Software Engineering Intern — *May 2025 – July 2025*
- Architected an agentic AI prompt-validation system using **Google ADK, Python, and FastAPI**, cutting latency by **40%** and delivering **98% accuracy**; deployed on **GCP** across Ford's global engineering teams, directly improving developer UX at scale.
- Built an automated quiz assessment module with a personalized STAR-method feedback engine, scaling structured evaluation to **200+ users** and boosting candidate engagement by **25%**.

Projects

DeepFake Voice Detection System | *Python, TensorFlow, Autoencoder, ResNet-18, Librosa, Streamlit*
- Designed a two-stage detection pipeline (Autoencoder anomaly scoring + ResNet-18 binary classification) targeting AI-generated audio fraud; trained on ASVspoof 2021 DF dataset and attained **92% accuracy** across unseen attack variants.

Pan-Cancer Detection & Classification | *Python, PyTorch, EfficientNet-B0, Vision Transformer, OpenCV*
- Built a hybrid CNN-Transformer model classifying **6 cancer types** from histopathological images; recorded **93.05% accuracy** overall and **93.94%** on sub-classification, matching clinical-grade benchmarks.

Travel Agency Booking SPA | *ReactJS, JavaScript, Material-UI, Vite, Firebase*
- Independently built and deployed a production travel booking platform featuring JWT auth, role-based routing, real-time filtering, and Vite code-splitting; reached **50+ users and 100+ sessions** with zero marketing spend.

Real-Time VoIP System | *Java, UDP Sockets, Maven*
- Implemented a dependency-free Java VoIP system supporting **4 concurrent peers** with custom UDP packet handling and jitter buffering, sustaining **sub-50 ms latency**; Maven-packaged for cross-platform deployment.

Technical Skills

Languages: Python, Java, C, JavaScript
Frontend: ReactJS, HTML5, CSS3, Vite
Backend / APIs: FastAPI, REST APIs, Google ADK
AI / ML: TensorFlow, PyTorch, OpenCV, Scikit-learn
Databases: MySQL, Oracle SQL
Tools & Practices: Git, GitHub, Firebase, Maven, Agile/Scrum

Certifications

Microsoft Certified: Azure Database Administrator Associate – **Microsoft PearsonVue**
Programming, DSA using Python – **IIT Madras (NPTEL)**
Python for Data Science – **IIT Madras (NPTEL)**
Career Essentials in Software Development – **Microsoft & LinkedIn**